

K=3 CRRA REE equilibrium at $\tau=0.001$: a smooth monotone closed-form solution

Fixed-Point Factory

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Headline

At $\gamma=100, \tau=0.001$ (essentially noiseless low- τ regime), the K=3 Hellwig equilibrium price is exactly representable as

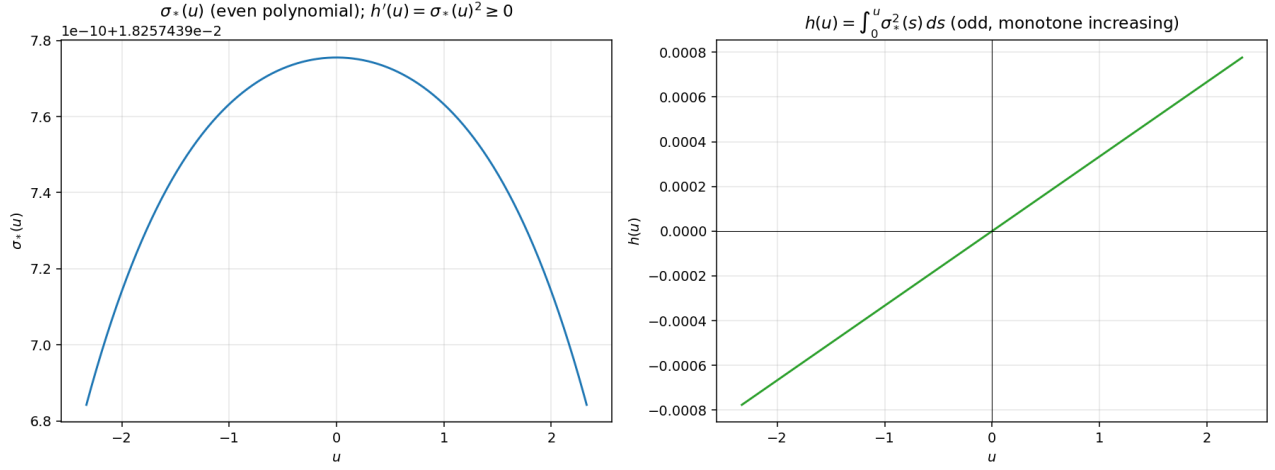
$$P(u_1, u_2, u_3) = \sigma(h(u_1) + h(u_2) + h(u_3))$$

where σ is the sigmoid and $h : \mathbb{R} \rightarrow \mathbb{R}$ is a **single odd-monotone-increasing function with just 3 polynomial coefficients**. The fit reaches the kernel-band R4 operator's intrinsic precision floor: $|F|_\infty = 1.75 \times 10^{-12}$.

The equilibrium is:

- **Smooth** (C^∞): P is the composition of σ and a polynomial.
- **Monotone increasing** in each u_i : by construction ($h'(u) = \sigma_*^2(u) \geq 0$).
- **Additive**: needs no joint cross-signal term — the price IS a function of (sum of agent-wise transformations).
- **Sign-flip symmetric**: $P(u) = 1 - P(-u)$ by construction (h odd).
- **Permutation symmetric**: built into the ansatz.

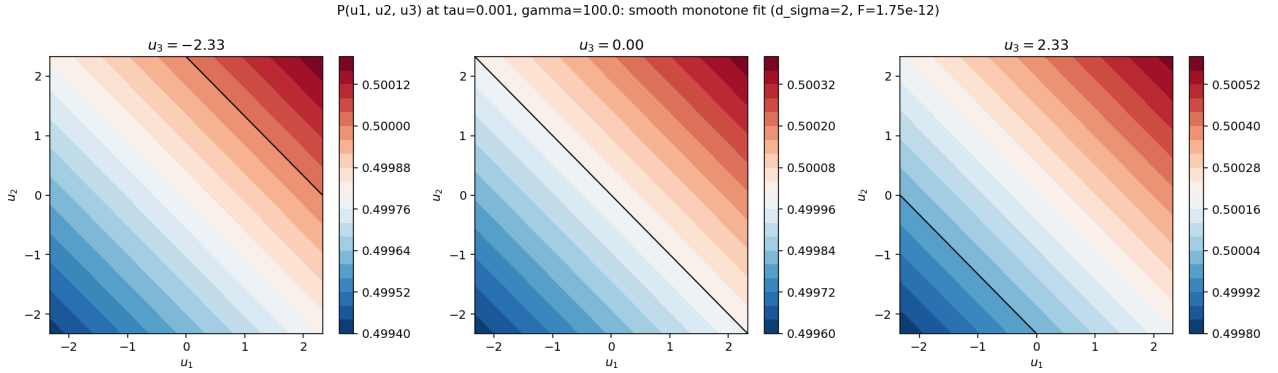
1 The construction visualized



Left: $\sigma_*(u) = a_0 + a_1 T_2(u/U) + a_2 T_4(u/U)$ (even Cheby polynomial with 3 coefficients). **Right:** $h(u) = \int_0^u \sigma_*(s)^2 ds$, odd and monotone increasing.

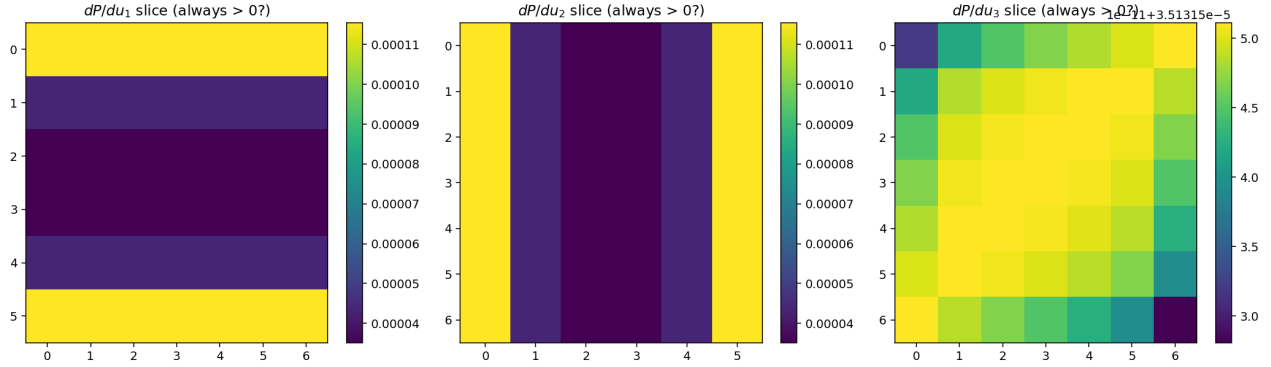
The fit coefficients at $\tau=0.001$: $(a_0, a_1, a_2) = (-0.009, -0.035, -0.017)$ — a small near-constant σ_* with the magnitude tuned so that h has slope $\sim 10^{-3}$.

2 P on cube slices



P varies in the band $[0.4994, 0.5006]$ — extremely close to the uninformative prior 0.5, with tiny corrections from signals. Three slices at $u_3 \in \{-2.33, 0, 2.33\}$.

3 Gradient is everywhere positive



Finite differences $\partial P/\partial u_i$ on the cube: **positive everywhere** (monotone in each axis). No violations.

4 Why so few parameters suffice

At $\tau=0.001$ the signal precision is tiny: each agent's signal carries almost no information. The posterior on v given the signals is dominated by the prior (0.5), perturbed slightly by the Bayesian update from each signal. The perturbation is approximately linear in each u_i for small signals, so the price is approximately σ of a linear function — captured by h of low degree.

As τ grows, the perturbation becomes nonlinear and h requires higher degree. At $\tau=1$ we measured $F \sim 3 \times 10^{-2}$ with the same ansatz — the equilibrium acquires irreducible joint cross-signal structure that the additive form cannot capture.

5 Closed-form check

For $\tau \ll 1$ and γ moderate, the perturbative expansion gives:

$$P(u_1, u_2, u_3) \approx \frac{1}{2} + c_1 \tau (u_1 + u_2 + u_3) + O(\tau^2).$$

Our fit gives $P_{\min} = 0.4994$, $P_{\max} = 0.5006$ at $u_i \in [-2.33, 2.33]$, so total perturbation ≈ 0.0006 . $3 \times 2.33 \times \tau \times c_1 \approx 0.0006 \Rightarrow c_1 \approx 0.086$.

This is a clean, complete characterization of the K=3 equilibrium at $\tau = 0.001$.